Assembly Language for x86 Processors
7th Edition, Global Edition

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Chapter 8: Advanced Procedures

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Chapter Overview

- **Stack Frames**
- Recursion
- INVOKE, ADDR, PROC, and PROTO
- Creating Multimodule Programs

Stack Frames

- Stack Parameters
- Local Variables
- ENTER and LEAVE Instructions
- LOCAL Directive
- WriteStackFrame Procedure

Stack Frame

- Also known as an *activation record*
- Area of the stack set aside for a procedure's return address, passed parameters, saved registers, and local variables
- Created by the following steps:
  - Calling program pushes arguments on the stack and calls the procedure.
  - The called procedure pushes EBP on the stack, and sets EBP to ESP.
  - If local variables are needed, a constant is subtracted from ESP to make room on the stack.

Stack Parameters

- More convenient than register parameters

```
pushad
mov esi,OFFSET array
mov ecx,LENGTHOF array
mov ebx,TYPE array

popad
```

```
push TYPE array
push LENGTHOF array
push OFFSET array
```

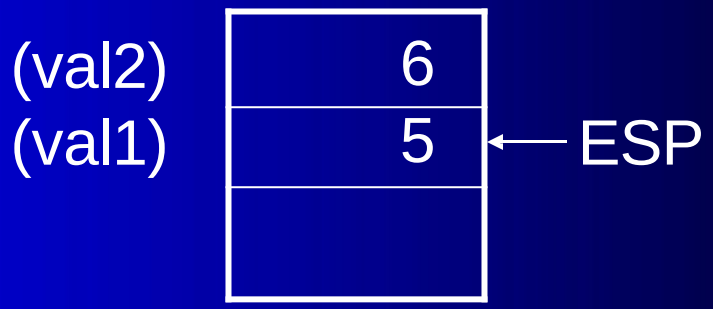
Passing Arguments by Value

- Push argument values on stack
- Call the called-procedure
- Remove arguments from the stack if the called-procedure did not remove them

Example

```
.data
val1   DWORD 5
val2   DWORD 6

.code
push val2
push val1
```



Stack prior to CALL

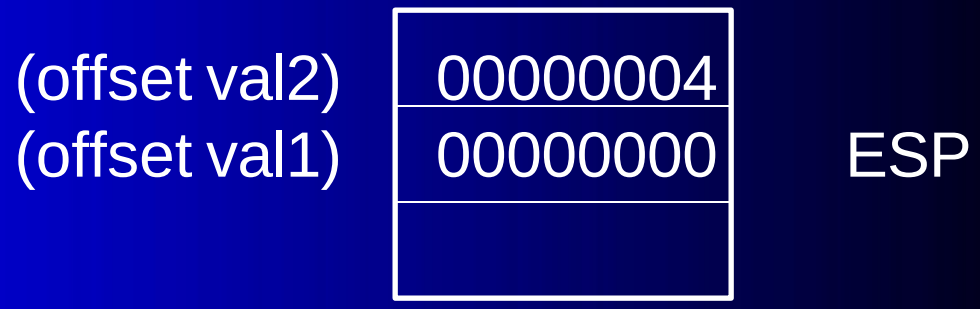
Passing by Reference

- Push the offsets of arguments on the stack
- Call the procedure
- Remove arguments from the stack if the called procedure did not remove them

Example

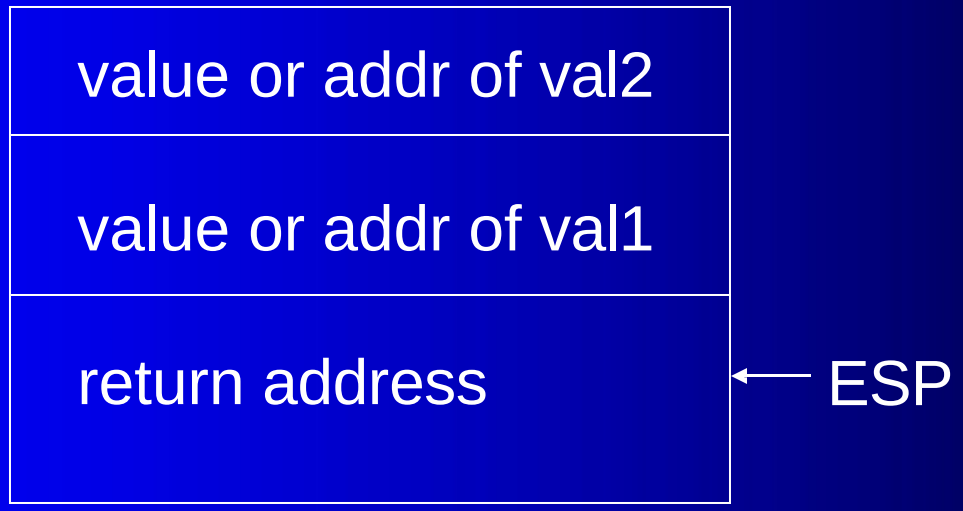
```
.data
val1   DWORD 5
val2   DWORD 6

.code
push OFFSET val2
push OFFSET val1
```



Stack prior to CALL

Stack after the CALL



Passing an Array by Reference (1 of 2)

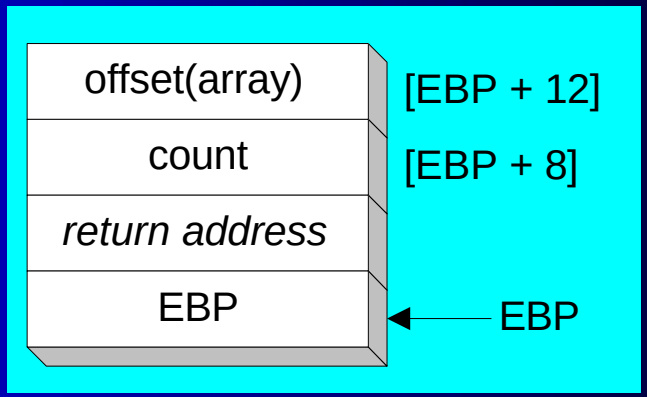
- The ArrayFill procedure fills an array with 16-bit random integers
- The calling program passes the address of the array, along with a count of the number of array elements:

```
.data
count = 100
array WORD count DUP(?)
.code
    push OFFSET array
    push COUNT
    call ArrayFill
```

Passing an Array by Reference (2 of 2)

ArrayFill can reference an array without knowing the array's name:

```
ArrayFill PROC
    push ebp
    mov  ebp,esp
    pushad
    mov  esi,[ebp+12]
    mov  ecx,[ebp+8]
    .
    .
```



ESI points to the beginning of the array, so it's easy to use a loop to access each array element. View the complete program.

Accessing Stack Parameters

- Functions access stack parameters using constant offsets from EBP.
  - Example: [ebp + 8]
- EBP is called the base pointer or frame pointer because it holds the base address of the stack frame.
- EBP does not change value during the function.
- EBP must be restored to its original value when a function returns.